

GLA UNIVERSITY, MATHURA

B.Tech. II Year, III/IV Semester

Subject Name-Probability and Statistics

Assignment-02

- 1) The probability of a man hitting the target is $\frac{1}{4}$.
 - (i) If he fires 7 times, what is the probability that he hits the target:
 - (a) at least once,
 - (b) at least twice
 - (ii) How many times must be fired so that the probability of his hitting the target at least once is greater than $\frac{2}{3}$?
- 2) A light bulb manufacturing factory finds 3 in every 60 light bulbs defective. What is the probability that the first defective light bulb will be found when the 6th one is tested?
- 3) A manufacturer who produces medicine bottles finds that 0.1% of the bottles are defective. The bottles are packed in boxes, each containing 500 bottles. A drug manufacturer buys 100 boxes from the producer of bottles. Using Poisson distribution, find how many boxes will contain (i) no defective bottles, (ii) at least two defective bottles. (given $e^{-0.5} = 0.6065$).
- 4) If a random variable X follows a Poisson distribution such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, Find the mean and variance of X.
- 5) In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error.
- 6) The marks obtained in a certain examination follow normal distribution with mean 45 and standard deviation 10. If 1000 students appeared at the examination, calculate the number of students scoring
 - (i) Less than 40 marks,
 - (ii) More than 60 marks.
- 7) The mean yield for one-acre plot is 662 kilos with a s.d. 32 kilos. Assuming normal distribution, how many one-acre plots in batch of 1000 plots would you expect to have yield (i) over 700 kilos, (ii) below 650 kilos, and (iii) what is the lowest yield of the best 100 plots?
- 8) Calculate the coefficient of correlation between the following data:

X:	10,000	20,000	30,000	40,000	50,000	60,000	70,000
Y:	0.3	0.5	0.6	0.8	1.0	1.1	1.3

9) The following table gives the ages and blood pressure of 10 women:

Age(X)	:	56	42	36	47	49	42	60	72	63	55
Blood Pressure(Y)	:	147	125	118	128	145	140	155	160	149	150

- (i) Find the correlation coefficient between X and Y.
- (ii) Determine the regression equation of Y on X.
- (iii) Estimate the blood pressure of a woman whose age is 45 years.

10) Regression equations of two variables X and Y are as follows:

$$8X - 10Y = 64 \quad \text{and} \quad 40X - 18Y = 320$$

Find (i) The means, (ii) The regression coefficients, and (iii) The coefficients of correlation between X and Y.

11) A die is tossed 120 times and each outcome is recorded as under:

Faces	:	1	2	3	4	5	6
Frequency	:	20	22	17	18	19	24

Is the distribution of outcomes uniform?

12) A certain drug is claimed to be effective in curing colds. In an experiment on 500 persons with cold, half of them were given the drug and half of them were given sugar pills. The patients reactions to the treatment are recorded in the following table:

Treatment	Helped	Reaction	No effect	Total
Drug	150	30	70	250
Sugar Pills	130	40	80	250
Total	280	70	150	500

On the basis of data, can it be concluded that there is a significance difference in the effect of the drug and sugar pills?